

MA 301
Test 1
Fall - 2013

Problem	points	out of
1		12
2		5
3		5
4		5
5		5
6		7
7		4
8		4
Score		47

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

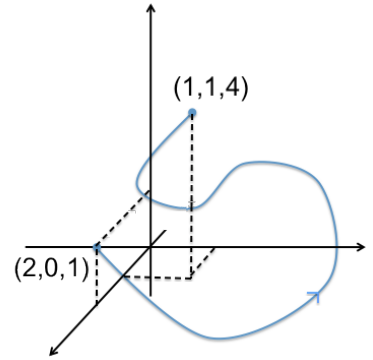
Problem 1: Compute $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$ for each of the following:

(a) $\mathbf{F} = [x^2 + 4y, x + y^2]$ and C is the rectangle in the xy -plane with $0 \leq x \leq 4$ and $-1 \leq y \leq 2$

(b) $\mathbf{F} = [y, x]$ and C is the parameterized path $x = t^3, y = t^2, 1 \leq t \leq 4$.

(c) $\mathbf{F}(x, y) = [2y, 6x]$ and C is the positively oriented circle $x^2 + y^2 = 100$.

Problem 2: Let $\mathbf{F} = \left[\frac{y^2}{x}, 2y \ln x, 3z^2 \right]$ compute the work required to move an object along C (shown below). Justify your work.



Problem 3: Suppose a straight wire from initial point $(0, 1, 5)$ to end point $(2, 2, 3)$ has a linear density function given by $\delta(x, y, z) = x^3 z$. Find the total mass of the wire.

Problem 4: Let $\mathbf{F} = \nabla(2x^2 + 2y^3)$ and C is the quarter of the circle $x^2 + y^2 = 1$ in the fourth quadrant, oriented counterclockwise. Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

Problem 5: Let C be the triangle with vertices $(0,0)$, $(1,0)$, $(1,4)$ oriented counter-clockwise and

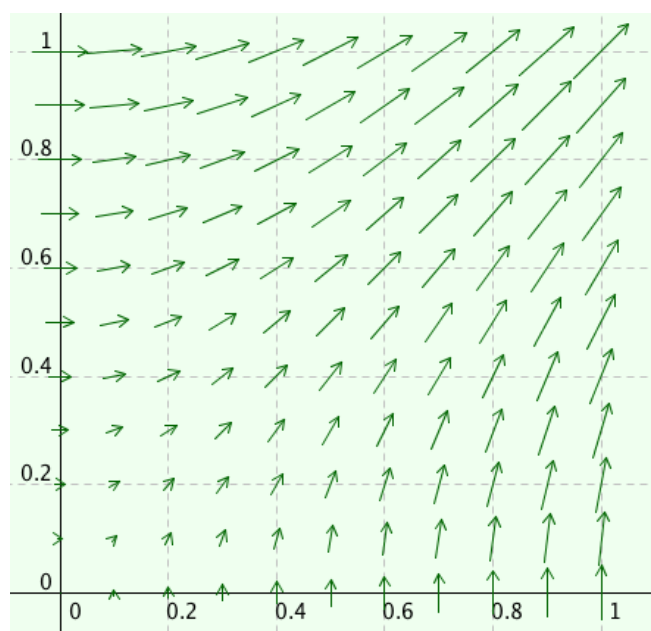
$$\mathbf{F}(x,y) = (xe^y - 3xy)\mathbf{i} + \frac{3}{2}y^2\mathbf{j}$$

Find the flux of $\mathbf{F}$ around C : $\int_C \mathbf{F} \cdot \mathbf{N} ds$ where $\mathbf{N}$ is the unit normal to C

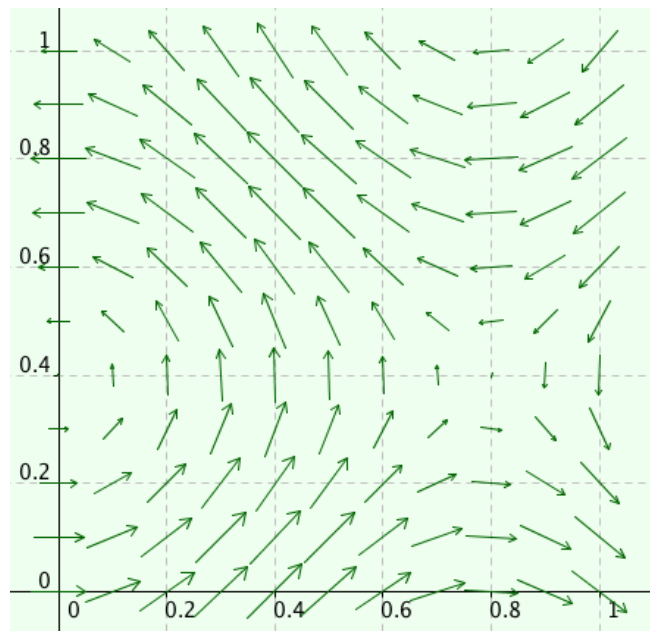
Problem 6: Answer the following true/false questions

T / F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
T / F	If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \neq 0$ for some closed path C , then $\mathbf{F}$ is path-dependent.
T / F	If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
T / F	If $\mathbf{F}$ is incompressible then $\mathbf{F}$ is conservative.
T / F	If $\mathbf{F}$ is conservative then the work required to move an object along any path is zero.
T / F	A vector field, $\mathbf{F}$, can be both incompressible and irrotational.
T / F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$

Problem 7: Consider the vector fields below. Based on the visual inspection, circle the terms below each figure that apply to that particular field.

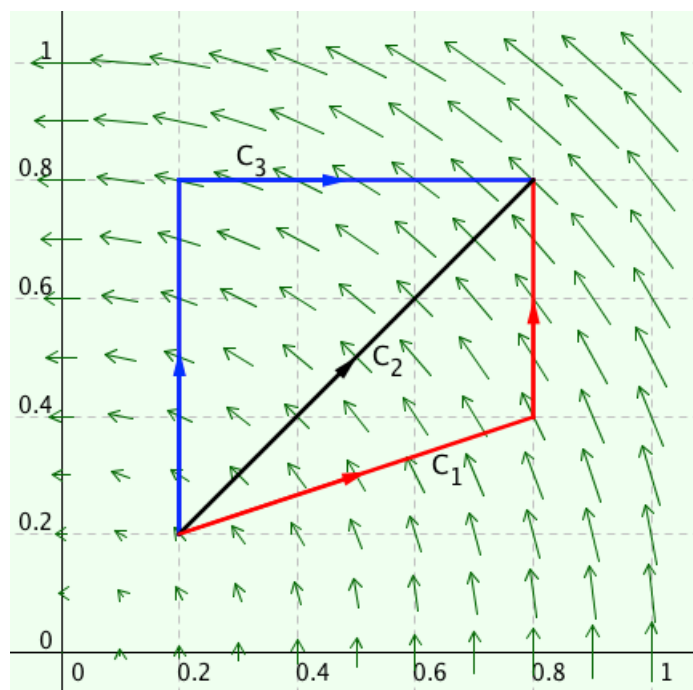


incompressible ** irrotational ** conservative



incompressible ** irrotational ** conservative

Problem 8: Consider the vector field $\mathbf{F}$ shown in the figure, together with the paths C_1 , C_2 , and C_3 . Order the line integrals below from smallest to largest. For full credit you must clearly explain your answer.



[1] $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$

[2] $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$

[3] $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$